Reg. No. | | | | | | | | | | [18MTCSE202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

ADVANCED SOFTWARE ENGINEERING

Time: 03 Hours **Maximum Marks: 100**

Instructions:
i. **Answer FIVE full Questions. Question 1 and 2 are compulsory.**
ii. **Assume to provide diagrams, example and illustrations wherever you feel relevant.**

1	a.	Compare Waterfall and Incremental development model of software development, providing a neat diagram.	10
	b.	Write short notes on Professional ethical responsibility of a software developer.	10
2.	a.	What are design patterns? Define the essential elements of any two design patterns with relevant examples.	10
	b.	Explain with a neat diagram, how various practices of Extreme Programming (Xp) support the agile philosophy (i.e. agile manifesto).	10
3.	a.	What is Component-based Software Engineering (CBSE)? Explain its essential components, advantage and limitations.	10
	b.	With a diagram explain in detail the system evolution process, giving relevant examples.	10
		OR	
4.	a.	Mention some of the strategic options available for the assessment of legacy system evolution. When would you normally replace rather than continue maintenance of the software (with or without re-engineering).	10
	b.	Briefly explain the three types of software maintenance. Why is it sometimes difficult to distinguish between them?	10
5.	a.	Define a critical system providing an example. Provide three types of critical systems with examples.	10
	b.	Explain in brief the various Security-requirements that may be included in a system.	10
		OR	
6.	a.	Describe in detail the software inspection process for the controller of an insulin pump.	10
	b.	What is regression testing? Explain how the use of automated tests and testing framework such as JUnit helps in agile methodology (e.g. Xp).	10

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7. a. Explain with an example how redundancy and diversity are fundamental strategies for enhancing the dependability of a system. **10**

b. Explain the process of exception handling giving example. Why you should explicitly handle all exceptions in a system that is intended to have a high level of availability. **10**

OR

8. a. Briefly explain the various steps in Security assessment with relevant examples; in addition mention typical items in a security checklist. **10**

b. Assume you are developing an innovative insulin pump which increases the efficiency of controlling diabetics by 90%. Explain why is it important for you to get FDA approval, if you want to launch this product in USA? And what steps would you follow to meet the compliance by FDA. **10**